



Predictive Credit Risk Pipeline

End-to-End Machine Learning Engineering

By: Lin Xiongqing

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Business Problem

Financial institutions need to **assess loan default risk** at the point of application. Traditional credit-scoring methods are **static and manual**, causing delayed or inconsistent risk evaluation. A **machine-learning-driven approach** enables faster, data-driven, and adaptive credit decisions.

Objective:

- Develop a **fully automated ML pipeline** that predicts **loan default probability** at the time of loan application

Key Highlights:

- End-to-end **MLOps lifecycle**: from raw data ingestion to model governance
- Modular, **Airflow-orchestrated** data pipeline
- **Versioned model bank** with drift detection and retraining trigger





Pipeline Architecture Overview

Data Pipeline

- Bronze → Silver → Gold
- Ingest, clean, and transform data into monthly Label & Feature Stores
- Enforces temporal joins to prevent leakage

Auto ML Pipeline

- Trains and evaluates XGBoost models
- Stores best model artefacts in a versioned Model Bank

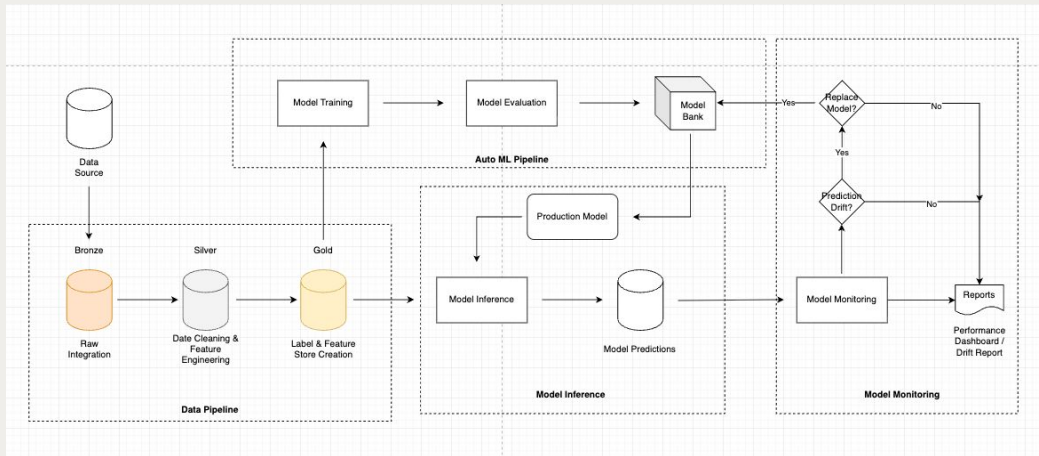
Model Inference

- Loads the latest production model
- Scores new applications and stores predictions

Model Monitoring

- Tracks AUC and default trends monthly
- Detects drift (>5% drop) and triggers retraining

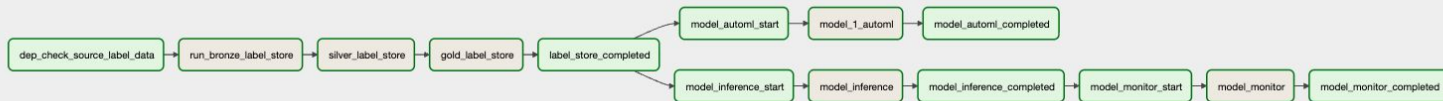
Overview: The pipeline automates the entire ML lifecycle—from data ingestion to model monitoring—fully orchestrated through **Apache Airflow** on a monthly schedule.





Airflow DAG Orchestration

Overview: The entire ML pipeline is automated and scheduled through an **Apache Airflow DAG**, ensuring monthly execution, version control, and reproducibility



Key Design Highlights:

- **DAG Name:** credit_risk_ml_pipeline
- **Schedule:** Runs **monthly** (0 0 1 * *) with **backfill enabled**
- **Resilience:** retries=3 with retry_delay=5 min
- **Structure:** Independent task groups for data, model, and monitoring stages.

Key Benefit:

- **Fully automated:** No manual trigger needed after setup
- **Modular and re-runnable:** Each stage can be re-executed independently
- **Backfilling support:** Enables time-based reprocessing for historical data
- **Governed workflow:** Ensures traceable execution across all pipeline stages





Data Pipeline (Medallion)

Bronze Layer – Raw Integration

- Ingests raw CSV data from source systems (loans, clickstream, attributes, financials)
- Preserves original schema and snapshot structure (YYYY-MM-DD)
- Stored in `/datamart/bronze/` as versioned CSVs
- **Purpose:** Maintain unaltered historical records for traceability

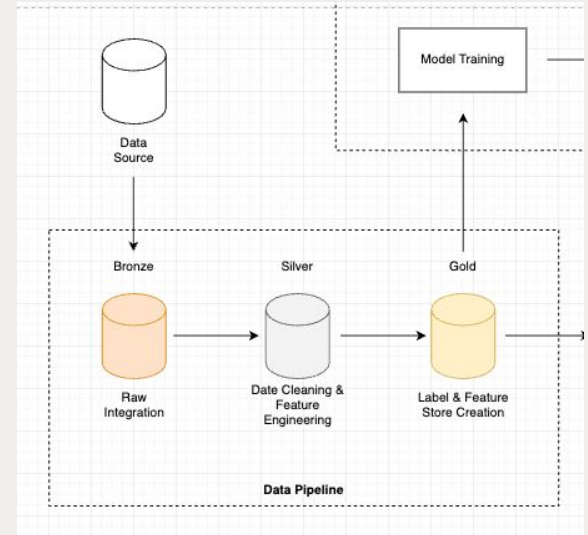
Silver Layer – Data Cleaning & Feature Engineering

- Enforces schema and type consistency using Spark
- Removes invalid entries (e.g., negative values, out-of-range ages)
- Handles missing data, standardizes text fields, and derives behavioral metrics (e.g., DPD, MOB)
- Stored in `/datamart/silver/` as Parquet for performance
- **Purpose:** Ensure high-quality, analytics-ready features

Gold Layer – Label & Feature Store Creation

- Combines multiple Silver datasets into training-ready tables
- Creates Label Store (target variable: 30+ DPD @ 6 MOB)
- Builds Feature Store via temporal joins (ensures no future leakage)
- Stored in `/datamart/gold/`
- **Purpose:** Provide consistent, time-aligned datasets for model training and inference

Overview: A layered data architecture ensures clean data lineage, temporal consistency, and prevents leakage throughout the ML lifecycle.



Reference:

<https://www.databricks.com/glossary/medallion-architecture>





Model Training Process (AutoML)

Training Setup

- **Algorithm:**
 - **XGBoost Classifier** with **RandomizedSearchCV** (100 iterations, 3-fold CV)
- **Data split** by time windows:
 - **Train/Test:** past 12 months
 - **OOT:** last 2 months (future simulation)
- **Target:** label = 1 if Days Past Due $\geq$ 30 at Month-on-Book = 6
- **Features:** Behavioral, financial, and demographic variables from Gold Feature Store

Preprocessing:

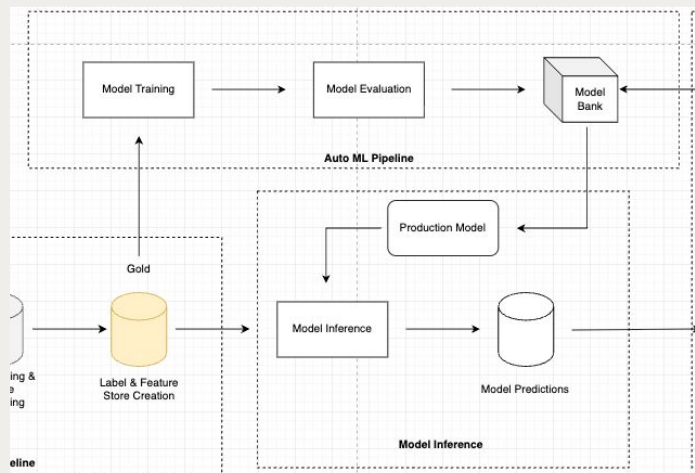
- Encodes categorical variables using category codes
- Replaces missing/ ∞ values with column means
- Scales numeric features using **StandardScaler** (fit on train only)
- Stores all preprocessing objects inside model artefact

Evaluation Metrics:

- **AUC/Gini** ensures balanced evaluation across time splits

```
'results': {'auc_oot': np.float64(0.8026369426751592),  
            'auc_test': np.float64(0.7846195631086796),  
            'auc_train': np.float64(0.9124923106907283),  
            'gini_oot': np.float64(0.605),  
            'gini_test': np.float64(0.569),  
            'gini_train': np.float64(0.825)}}
```

Overview: The **AutoML module** trains, evaluates, and version-controls the best-performing model using historical data snapshots from the Gold layer.





Model Inference (Scoring)

1. Load model artefact

- Retrieves model, scaler, and metadata from `model_bank/auto_ml/`
- Example: `credit_model_2024_09_01.pkl`

2. Load current features

- Reads snapshot from `datamart/gold/feature_store/` for the selected month
- Ensures only data $\leq$ snapshot date is used (no leakage)

3. Preprocess for inference

- Reuses stored **StandardScaler** and categorical mappings
- Handles missing values and invalid entries

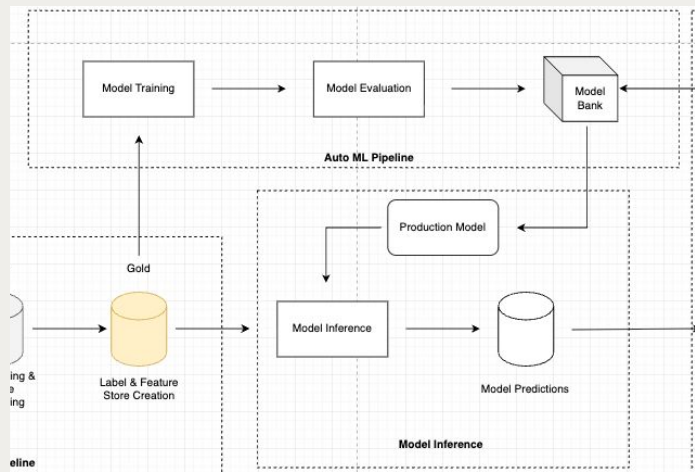
4. Generate predictions

- Computes probability of default:
- `'Model.predict_proba(X)[:,1]' → score between 0–1`

5. Store results

- Saves outputs to `datamart/gold/model_predictions/credit_model_YYYY_MM_DD/`
- File name pattern: `credit_model_YYYY_MM_DD_predictions_YYYY_MM_DD.parquet`

Overview: The inference stage loads the **production trained model** from the **Model Bank**, applies the same preprocessing pipeline, and generates **monthly loan default probabilities** for all active customers.





Monitoring & Drift Detection

1. Load inputs

- Reads predictions from `datamart/gold/model_predictions/`
- Loads actual labels from `datamart/gold/label_store/`

2. Merge & evaluate

- Joins by **Customer_ID**
- Calculates **AUC** for current snapshot vs model's training OOT AUC

3. Drift detection logic

- $AUC_{diff} = latest_AUC - baseline_AUC$
- If $AUC_{diff} < -0.05 \rightarrow$ drift detected
- Creates **model_drift_detected.flag** to trigger retraining

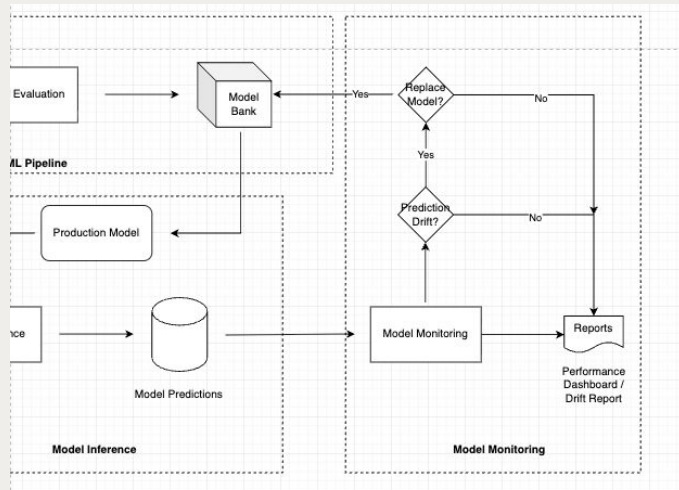
4. Historical tracking

- Aggregates monthly metrics:
AUC, Default Rate, Mean Prediction, Std, Record Count
- Saves summary to `datamart/gold/model_monitor_summary/`

5. Visualization

- Plots AUC vs Default Rate trend over time
- Stored in `reports/model_perf_trend_YYYY_MM_DD.png`

Overview: The monitoring stage evaluates deployed model performance over time, detects drift, and triggers retraining when performance falls below threshold.





Model Governance & SOP

1. Version Control

- Each trained model is saved as a versioned artefact:
`credit_model_YYYY_MM_DD.pkl`
- Artefact stores model, preprocessing objects, metadata, and training results

2. Performance Monitoring

- Model AUC and Default Rate tracked monthly using predictions and actuals
- Results saved to `/datamart/gold/model_monitor_summary/`

3. Retraining Policy

- Trigger:** AUC drop > 5% from model's training OOT AUC
- When triggered, Airflow creates `model_drift_detected.flag` → initiates retraining in next DAG cycle

4. Data Integrity & Auditability

- All datasets snapshot-based (YYYY-MM-DD) → reproducible experiments
- Historical versions preserved for audit and validation

Overview: Model governance ensures every trained model is **traceable, monitored, and refreshed** based on performance stability and business risk tolerance.

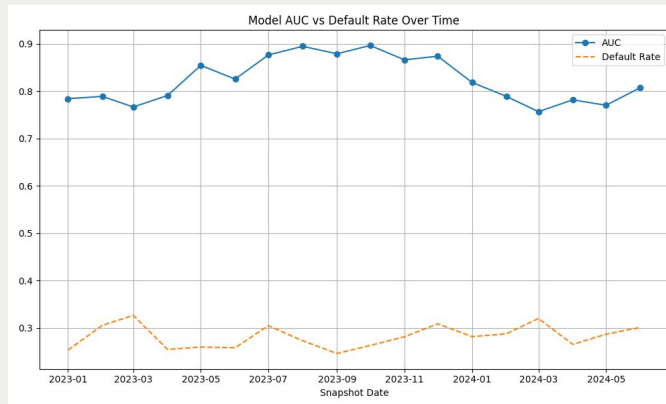


Figure: Model AUC (blue) remains stable across months while Default Rate (orange) fluctuates within expected range.

Drop in AUC beyond -0.05 indicates model drift, prompting retraining under governance SOP.





Business Impact & Future Enhancements

Current Impact

- **Faster Credit Assessment:** End-to-end scoring automated monthly via Airflow
- **Data-Driven Risk Control:** Accurate, explainable default probability predictions
- **Model Governance:** Monitored AUC stability and retrain triggers prevent performance decay

Future Enhancements

- **Feature Drift Monitoring:** Extend monitoring to feature distribution shifts (e.g., KS tests)
- **MLflow / Metadata Tracking:** Centralize experiment tracking and artefact lineage
- **Deployment Integration:** Expose inference via API or batch service endpoint
- **Dashboard Automation:** Auto-sync model reports into BI dashboards for stakeholders

Outcome

- A scalable, production-ready ML pipeline that ensures **operational efficiency, model transparency, and regulatory compliance** in credit risk decisioning.

